Do Post-Stroke Patients Experience Less Fatigue after tDCS Therapy: A Systematic Review and Meta-Analysis

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Abstract

Objective Post-stroke patients experience a lot of different symptoms. Brain stimulation has already been proven to be effective for symptoms like chronic pain and depression. Lots of patients experience fatigue, but the effect of tDCS on fatigue is still unclear.

Methods A systematic review was conducted for all published studies untill April 2024 using PubMed. Studies were excluded if they had wrong design, did not use tDCS as intervention or did not have fatigue measured before and after the intervention. Two reviewers screened and assessed the relevant studies for eligibility. One reviewer did a quality assessment which was considered in the meta-analysis using the Cochrane tool.

Results Four meta-analyses were performed. One about effects between the difference in fatigue scores before and after tDCS or before and after sham-tDCS. The other one describes fatigue scores pre- and post-tDCS. Both can be divided into short-term effects and long-term effects.

Conclusion tDCS compared to sham procedure shows no effect on short- and long-term fatigue. However, tDCS is a yet to be fully explored treatment for post-stroke fatigue. Its role in rehabilitation should be further developed and applied more broadly within various therapies to gain a deeper understanding of its uses.

Keywords stroke, transcranial direct current stimulation, fatigue.

1. Introduction

There are over 12.2 million new strokes each year (1). About one in two stroke survivors experience post-stroke fatigue (PSF) (2). It prevents patients from engaging in rehabilitation and other daily activities. Survivors indicated understanding fatigue and how to reduce it as one of the top research priorities to improve stroke outcomes (3). Current treatment for PSF mainly include pharmacological interventions, but more research is needed to determine optimal medication therapy (4). The medications that are commonly used in patients with stroke such as sedatives, antidepressants and hypnotics could also cause fatigue (3). The third Stroke Recovery and Rehabilitation Roundtable (SRRR3) states that neuromodulation therapies need to be prioritized for future research (5). Meanwhile rehabilitation guidelines are also starting to acknowledge the potential of brain stimulation for post-stroke recovery and recommends future research (6).

Transcranial direct current stimulation (tDCS) is a non-invasive brain stimulation (NIBS) method that applies a weak electrical current between 0.2-1.0 mA for several minutes to the scalp between an anodal (positive) and a cathodal (negative) electrode (7). This alters the resting membrane potential of neurons. Therefore, neurons under the anode undergo depolarization and are more likely produced by neuronal fire, while neurons under the cathode are hyperpolarized and are less likely to fire.

tDCS proved to be a useful treatment for chronic pain and depression (8), but the effect on PSF is currently not clear. The aim of the current review and meta-analysis is therefore to investigate the effect of tDCS on fatigue in stroke survivors.

2. Methods

2.1 Search strategy

2.1.1 Literature search

We searched in the database PubMed to find studies that reported fatigue in post-stroke patients and using tDCS as therapy. The following search terms were used: (transcranial direct current stimulation [MeSH] OR transcranial direct current stimulation OR tdcs OR Transcranial Electrical Stimulation) AND (stroke [MeSH] OR stroke) AND (fatigue [MeSH] OR fatigue OR post-stroke fatigue OR PSF). All studies before the 20th of April 2024 were included in this systematic review and meta-analysis.

2.1.2 Exclusion criteria

Studies were excluded if they were not research designs (e.g. protocols, reviews and collection of abstracts), did not use tDCS as intervention or did not have fatigue measured before and after the intervention.
For the measurement tool of fatigue, we included the Fatigue Severity Scale (FSS), the Visual Analogue Scale for Fatigue (VAS-F) and the Swallowing Quality of Life Questionnaire (SWAL-QoL). These questionnaires assigns a value to the intensity of the patients' fatigue. If an article had a different measurement tool it was also excluded. There were no restrictions on other patient characteristics as age and sex.

2.1.3 Selection procedure

For this study there were 2 independent reviewers screening the studies based on the eligibility criteria. If the reviewers JS and AR disagreed upon the studies, a third reviewer WG also assessed that study. Firstly, duplicates were removed. Secondly, titles and abstracts were screened and thirdly, full text screening was performed.

2.2 Quality assessment

We developed a quality assessment list based on VerHagen's Delphi list (9). It consists of 14 criteria based on current procedures for this therapy and expectations for robust research methodology relevant to our research question. Our criteria can be found in the appendix (table 1). These criteria are used to determine the quality of each paper, reflecting its validity and reliability. However, these criteria do not influence the inclusion or exclusion of papers in this systematic review. Our evaluation included verifying the use of appropriate randomization techniques to avoid selection bias and assessing the use of sham control groups. We also checked for the blinding of outcome assessors and care providers, which is crucial for minimizing bias in measurements, among other important aspects within our research question. This quality check aims to highlight

strengths and identify potential weaknesses, thereby projecting a clearer picture of the scientific rigor of the papers.
Each criterion was scored with a "yes," "doesn't say," or "no," corresponding to scores of 2, 1, and 0, respectively. We identified a study that scored between 19 and 28 points as a high-quality study, between 8 and 18 points as a medium quality study and between 0 and 7 points as a low-quality study.

2.3 Data extraction

We extracted data using standardize forms according to the PRISMA guidelines. The following data were extracted:

1. Characteristics of the included studies: authors, year of publication, institution, country of origin, study design (feasibility study, double-blind sham-controlled study, randomized sham-controlled study, randomized controlled trail, clinical case), male to female ratio, types of stroke, duration of data recruitment, inclusion criteria, primary and secondary outcome measure and the tool for the measurement of fatigue.
2. tDCS procedure and any other rehabilitation training used.
3. The fatigue scores from the baseline and after the intervention or sham.

2.4 Statistical analysis

Having retrieved the data, the Cochrane tool was used to compare the studies. The articles using different outcome measures were compared using standardized mean difference (SMD). The heterogeneity of the study was evaluated using Higgins inconsistency index (I^2), with substantial heterogeneity being indicated by an I^2 value greater than 50%, a random-effects model was used, otherwise the fixed-effects model was used. With this information 4 forest plots were made. Two comparing the difference in fatigue scores in the intervention and control groups if the study was an RCT and two comparing the fatigue scores pre- and post-tDCS. Both are subdivided in the effect on fatigue right after the procedure (short-term effect) and after more than 4 weeks (long-term effect). With the same data funnel plots were made. The Z- and p-value were also calculated using the confidence intervals, with $Z = 0$ meaning that there is no difference between the studies and $p < 0.05$ meaning that there is a statistically significant difference between the studies. If there was no mean or SD given, we calculated those ourselves if possible.
Our main outcome was comparing tDCS to the sham procedure, this in long- and short-term effects. As a secondary outcome we looked at the difference in scores pre- and post-tDCS divided into long- and short-term effects. All statistical analyses were conducted by one author: AR.

3. Results

3.1 Study selection

Our systematic literature search initially identified 34 articles from PubMed, as can be seen in figure 1. From the 34 articles one was removed as a duplicate. After this, all articles were screened based on title and abstract, resulting in 25 exclusions. Reasons for exclusion included articles being protocols, reviews, collections of

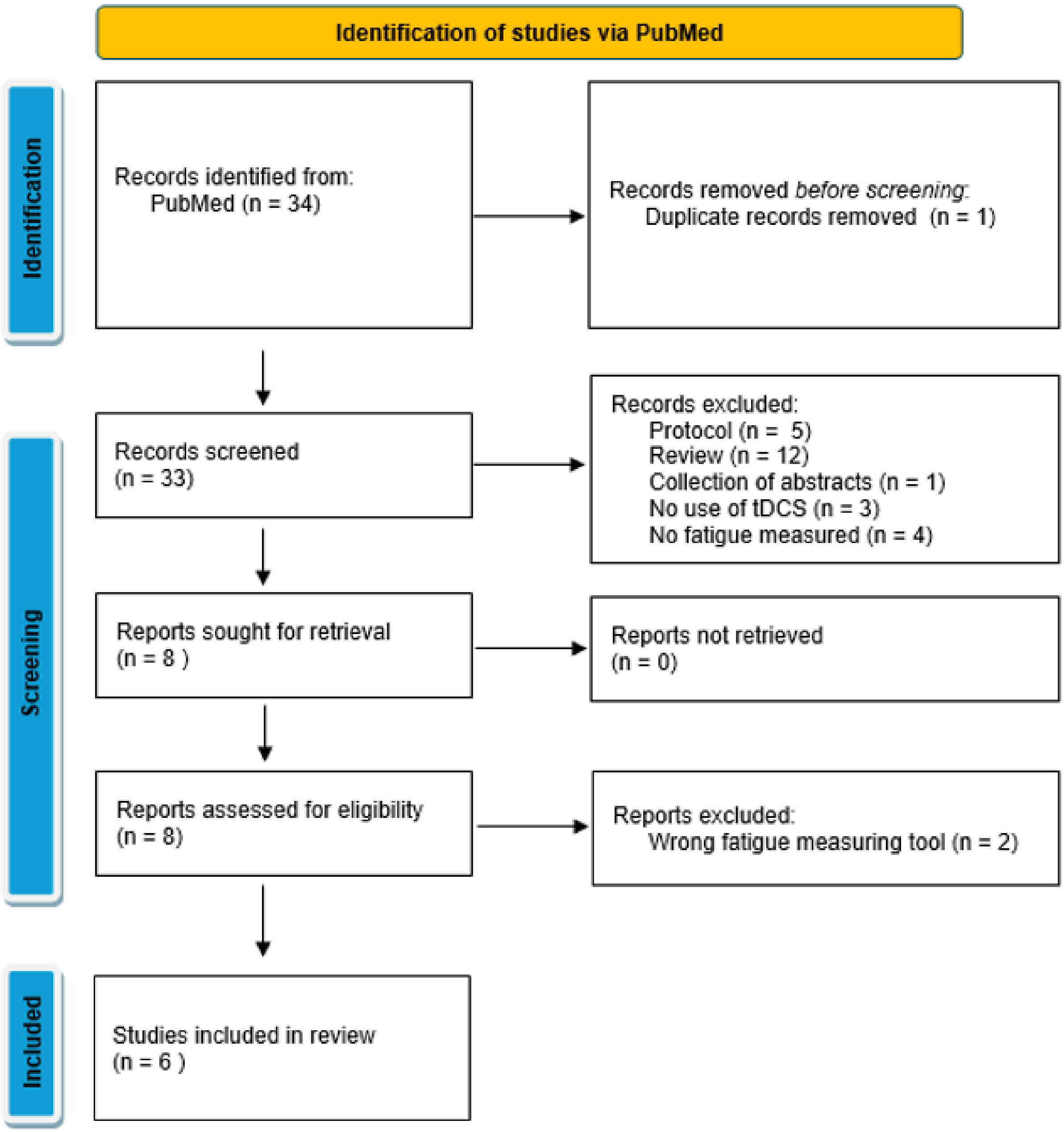

Figure 1. Flow chart

abstracts, no use of tDCS as intervention and no fatigue measured before and after the intervention. None of the articles were unable to be retrieved. After this screening, two more articles were excluded due to using a different measuring tool for fatigue, which were not FSS, VAS-F or SWAL-QoL. The remaining 6 articles were included in the meta-analysis.

3.2 Characteristics of the included studies

A summary of all the studies included can be found in the appendix (table 2). Most studies used a controlled trial. The total number of patients taken account in this systematic review is 211. Patients had multiple types of strokes, mostly chronic. The mean age of all the patients is 58.40 years with an SD of 10.89. 76% of all the patients were male. Most of the studies reported a follow-up of 5-6 weeks. The most used measurement scale for fatigue is the Fatigue Severity Scale (FSS).

3.3 Quality assessment and publication bias

Only Dong et al. (10) qualified as a high-quality study. De Doncker et al. (11), Sanchez-Kuhn et al. (12) and Ulrichsen et al. (13) qualified as a medium quality study. The other two studies (14, 15) were qualified as a low-quality study. All can be seen in the appendix in table 1. Not many studies used tDCS according to guidelines (duration of 30 minutes, repeated procedure 5 days a week for 3 weeks). Most studies were randomized.

We looked at the funnel plot, as can be seen in figure 2. It showed no sign of publication bias.

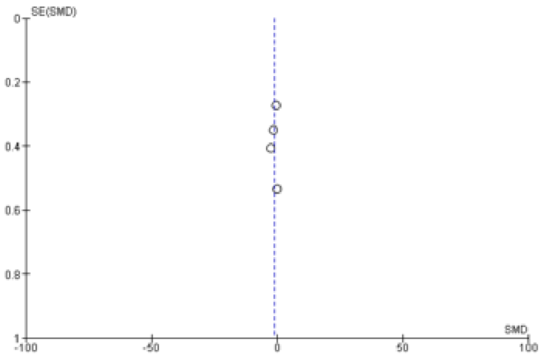

Figure 2. Funnel plot

3.4 Outcomes

For all the meta-analyses a random effects model was used, because every I^2 value was above 50% and therefore there was substantial heterogeneity.

The first meta-analysis can be found in figure 3. It describes the short-term effects between the difference in fatigue scores before and after tDCS or before and after sham-tDCS. The SMD is 0.64 [-0.44, 1.72] and with p = 0.25.

The second meta-analysis describes the long-term effect between the difference in fatigue scores before and after tDCS or before and after sham-tDCS, which can be found in figure 4. The SMD is 0.62 [-0.20, 1.43] and with p = 0.14.

The third meta-analysis can be found in figure 5. It describes the short-term effects on fatigue scores pre- and post-tDCS. The SMD is -1.77 [-3.18, -0.36] and with p = 0.01.

The fourth and last meta-analysis describes the long-term effects on fatigue scores pre- and post-tDCS, as can be seen in figure 6. The SMD is -1.04 [-2.13, 0.06] and with p = 0.06.

In the case study (12) there was a long-term enhancement of 20 points in the quality of life of the domain of fatigue.

4. Discussion

Our systematic review and meta-analys is determined the effectiveness of transcranial direct current stimulation (tDCS) on post-stroke fatigue to have an insignificant effect, compared to the sham-tDCS group.

However, the results from our meta-analysis show a statistically significant short-term reduction in fatigue due to the treatment when not compared to the sham procedure. This could be explained by the normal rehabilitation of

Short term Study or Subgroup	tDCS Mean	SD	Total	sham-tDCS Mean	SD	Total	Weight	Std. Mean Difference IV, Random, 95% CI
De Doncker et al. (2021)	1.42	0.63	20	0.35	0.74	10	30.9%	1.56 [0.69, 2.43]
Dong et al. (2021)	14.26	5.98	23	9.48	5.3	22	34.4%	0.83 [0.22, 1.44]
Gandiga et al. (2006)	0.14	0.28	23	0.25	0.3	23	34.7%	-0.37 [-0.96, 0.21]
Total (95% CI)			**66**			**55**	**100.0%**	**0.64 [-0.44, 1.72]**

Heterogeneity: Tau² = 0.79; Chi² = 15.38, df = 2 (P = 0.0005); I² = 87%
Test for overall effect: Z = 1.16 (P = 0.25)

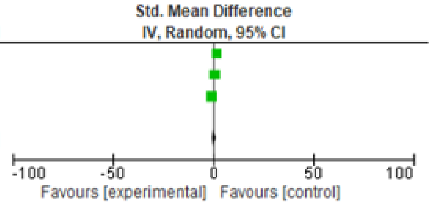

Figure 3. Forest plot: short term effects between the difference in fatigue scores before and after tDCS or before and after sham-tDCS

Long term Study or Subgroup	tDCS Mean	SD	Total	sham-tDCS Mean	SD	Total	Weight	Std. Mean Difference IV, Random, 95% CI
De Doncker et al. (2021)	0.61	0.63	20	-0.02	0.74	10	30.2%	0.92 [0.12, 1.72]
Dong et al. (2021)	15.03	5.98	23	8.61	5.3	22	33.9%	1.11 [0.48, 1.75]
Ulrichsen et al. (2021)	0.34	1.5	27	0.49	1.5	27	36.0%	-0.10 [-0.63, 0.44]
Total (95% CI)			**70**			**59**	**100.0%**	**0.62 [-0.20, 1.43]**

Heterogeneity: Tau² = 0.41; Chi² = 9.51, df = 2 (P = 0.009); I² = 79%
Test for overall effect: Z = 1.49 (P = 0.14)

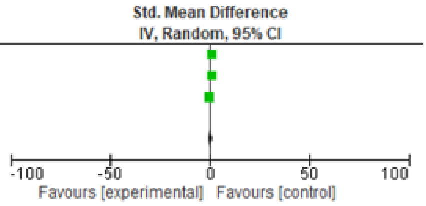

Figure 4. Forest plot: long term effects between the difference in fatigue scores before and after tDCS or before and after sham-tDCS

Short term Study or Subgroup	post-tDCS Mean	SD	Total	pre-tDCS Mean	SD	Total	Weight	Std. Mean Difference IV, Random, 95% CI
De Doncker et al. (2021)	4.42	0.32	20	5.84	0.63	20	32.0%	-2.79 [-3.68, -1.89]
Dong et al. (2021)	32.12	7.1	23	46.38	5.98	23	33.4%	-2.14 [-2.87, -1.40]
Gandiga et al. (2006)	7.81	0.29	23	7.95	0.28	23	34.6%	-0.48 [-1.07, 0.10]
Total (95% CI)			**66**			**66**	**100.0%**	**-1.77 [-3.18, -0.36]**

Heterogeneity: Tau² = 1.41; Chi² = 22.33, df = 2 (P < 0.0001); I² = 91%
Test for overall effect: Z = 2.47 (P = 0.01)

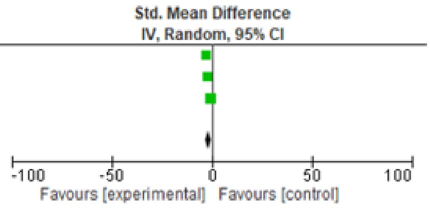

Figure 5. Forest plot: short term effects pre- and post-tDCS

Long term Study or Subgroup	post-tDCS Mean	SD	Total	pre-tDCS Mean	SD	Total	Weight	Std. Mean Difference IV, Random, 95% CI
Cleland et al. (2020)	70.8	24.7	7	71.8	17.3	7	22.7%	-0.04 [-1.09, 1.00]
De Doncker et al. (2021)	5.23	0.22	20	5.84	0.63	20	25.7%	-1.27 [-1.95, -0.58]
Dong et al. (2021)	31.35	5.47	23	46.38	5.98	23	24.8%	-2.58 [-3.38, -1.78]
Ulrichsen et al. (2021)	3.16	1.5	27	3.5	1.5	27	26.8%	-0.22 [-0.76, 0.31]
Total (95% CI)			**77**			**77**	**100.0%**	**-1.04 [-2.13, 0.06]**

Heterogeneity: Tau² = 1.10; Chi² = 26.73, df = 3 (P < 0.00001); I² = 89%
Test for overall effect: Z = 1.85 (P = 0.06)

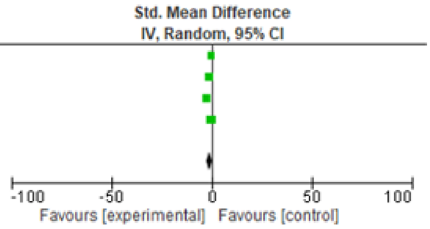

Figure 6. Forest plot: long term effects pre- and post-tDCS

fatigue overtime instead of the tDCS procedure, as could also be seen in the case study (12), but this was not considered clinically significant. This would explain the cause for no significant short-term effect between tDCS procedure and sham-tDCS.

Both meta-analysis about long-term effects were not significant. This means despite using tDCS no significant effect was found after 4 weeks of procedure. This indicates that fatigue does not decrease after 4 weeks.

The studies included in our analysis varied in quality, which we ultimately decided did not impact the weight of the studies within our meta-analysis. Gandiga et al. (14) was assessed as a low-quality study. However, this study stated a significant effect after tDCS compared to sham-tDCS unlike De Doncker et al. (11) and Dong et al. (10), which were rated as a high quality and medium quality study, respectively. This indicates that it's possible that low quality studies favor having an effect after tDCS procedure.

Further clinical implementation of tDCS should be considered with this study in mind. Not many studies could be included due to a lack of research in this field. Additionally, more research should be directed into the long-term effects of tDCS. It’s important to understand how much fatigue is reduced after 4 weeks and if it differs from using no tDCS.

In conclusion, tDCS compared to sham procedure shows no effect on short- and long-term fatigue. However, tDCS is a yet to be fully explored treatment for post-stroke fatigue. Its role in rehabilitation should be further developed and applied more broadly within various therapies to gain a deeper understanding of its uses.

Appendix

Table 1. Quality assessment of included studies

	Cleland et al. (2020) (12)	De Doncker et al. (2021) (11)	Dong et al. (2021) (10)	Gandiga et al. (2006) (13)	Sánchez-Kuhn et al. (2019) (14)	Ulrichsen et al. (2022) (15)
Was the method randomization performed	Doesn't say	Yes	Yes	Yes	Doesn't say	Yes
Was the study double-blinded	No	Yes	Yes	Doesn't say	Doesn't say	Doesn't say
Were the groups similar in baseline	Doesn't say	Yes	Yes	No	Doesn't say	Yes
Were the outcome assessors blinded	Doesn't say	Yes	Yes	No	Doesn't say	Doesn't say
Were the inclusion criteria clearly stated	Yes	Yes	Yes	No	Doesn't say	No
Were the exclusion criteria clearly stated	Yes	Yes	Yes	No	Doesn't say	No
Did they use the FSS score as part of an inclusion criterium	No	Yes	Yes	No	No	Yes
Was the number of patients > 40	No	No	Yes	No	No	Yes
Was there a dropout rate <15%	No	Yes	Yes	Doesn't say	Doesn't say	No
Was the patient followed up for at least 8 weeks	No	No	Yes	No	No	Yes
Was the duration of stimulation at least 30 minutes long	No	Yes	No	No	No	No
Was the procedure repeated at least 5 days a week for 3 weeks	No	No	Yes	No	No	Yes
Were other interventions used	No	No	Yes	No	Yes	No
Are the patients representative for the whole population	No	No	Yes	No	No	Yes
Total points	7	20	28	4	9	18

Table 2. Summary of included studies

	Affiliation	Study design	Number of patients	Stroke type	Age	Male %	Time of assessment	Fatigue measurement tool
Cleland et al. (2020) (12)	University of Illinois at Chicago	Feasibility study	7	Single unilateral stroke	59.7 (5.4)	85.7	5-6 weeks	VAS-F
De Doncker et al. (2021) (11)	University College London	Double-blind, sham-controlled study	30	All types	58.39 (12.42)	50.0	5 weeks	FSS-7
Dong et al. (2021) (10)	Nanchang University	Randomized controlled trial	53	All types	59.95 (9.44)	64.2	12 weeks	FSS
Gandiga et al. (2006) (13)	National Institutes of health, Bethesda University Teubingen	Double-blind, sham-controlled study	47	Chronic stroke	56.7 (5.2)	51.2	1 day	VAS 1-10
Sánchez-Kuhn et al. (2019) (14)	University of Almería	Clinical case	1	Severe dysphagic chronic cerebellar stroke	64	100	6 weeks	SWAL-QoL questionnaire
Ulrichsen et al. (2021) (15)	University of Oslo	Randomized sham-controlled trial	73	Chronic stroke	65.0 (12.2)	68.5	8 weeks	FSS

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